

Reflection: Value of the Agentic Test Case Generator

Capstone reflection for the Agentic AI Test Case Generator project.

The agent added the most value in coverage analysis. Manually writing test cases for straightforward acceptance criteria is possible, but it is easy to miss boundary conditions, edge cases, and cross-rule interactions. The generate-then-critique loop helped identify important checks such as account lockout after correct credentials are entered, session expiry after 24 hours, promo code replacement behavior, revalidation after cart changes, and discount capping so a subtotal never becomes negative.

The agent was also useful for consistency. It produced test cases in a structured format with traceability to acceptance criteria, priority, risk area, test data, steps, and expected results. This makes the output easier to import into tools such as TestRail or Zephyr and easier for reviewers to audit.

However, the agent was not equally valuable for every part of the work. Simple happy-path test cases, such as successful login or applying a valid promo code, could have been written manually just as quickly. The main benefit came from reviewing coverage, finding overlooked scenarios, and producing standardized outputs.

The agent needed guardrails. Without a critic step, it may miss acceptance criteria or create duplicate cases. The self-review step improves quality by checking every acceptance criterion and verifying that positive, negative, boundary, and edge categories are represented.

Overall, the agent genuinely adds value as a QA assistant rather than a replacement for a QA engineer. It accelerates test design, improves traceability, and reduces the chance of missing important scenarios. Human review is still needed to validate business assumptions, confirm test data, and align the suite with actual product behavior.